

DESIGN ISSUES:

1. The table named ORDER is a reserved SQL keyword, which forces every query and DDL statement referencing it to wrap the name in quotes and a persistent source of errors that is best avoided by renaming it to ORDERS, SALES_ORDER, or PURCHASE_ORDER.
2. Column names such as CSTMR, PRO, SID, OID, STR, NO, GEND, and BDAY are not self-documenting, reducing readability and making long-term maintenance significantly harder and they should be replaced with full descriptive names like CUSTOMER_ID, PRODUCT_ID, SORT_ID, BOTTLE_ID, STREET_ADDRESS, STREET_NUMBER, GENDER, and BIRTH_DATE.
3. The foreign key columns ORDER.CSTMR, ORDER.PRO, PRODUCT.SID, and PRODUCT.OID are all nullable, which undermines referential integrity by allowing orders to exist without an associated customer or product unless a specific business requirement justifies it, these columns should be marked NOT NULL.
4. PRODUCT.PRICE is defined as NUMBER(3,2), capping the maximum price at 9.99 and making it impossible to sell wines, spirits, or any product priced above that threshold and the column should be widened to at least NUMBER(7,2) or NUMBER(10,2).
5. ORDER.AMOUNT is defined as NUMBER(5,0), which limits order amounts to a maximum of 99,999 and may prove insufficient depending on actual business volume, the precision should be reviewed and increased to match real-world requirements.
6. A UNIQUE constraint has been placed on the DATE column, which means only one order can ever be recorded per day, this is a significant business limitation that should be corrected by either removing the constraint or changing it to UNIQUE(DATE, TIME).
7. CUSTOMER.GEND is stored as NUMBER(1,0) with no indication of what each numeric value represents, making the field unclear to anyone reading the schema, so it should be replaced with VARCHAR2(1) using explicit values such as 'M', 'F', and 'U' for readability and clarity.
8. All constraints follow the system-generated SYS_C0020XXX naming format, which is meaningless without querying the database to identify them and the meaningful names such as PK_BOTTLE, FK_PRODUCT_SORT, or UQ_ORDER_DATE should be used instead to improve documentation and maintainability.